A Symplectic Phase-Space Approach to Macroeconomics: Sectoral Balances, Geometric Torsion, and Systemic Ejections

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Abstract

Mainstream neoclassical economics traditionally models global markets as frictionless systems of infinite growth, failing to mathematically account for the systemic debt crises that periodically fracture the global economy. This paper introduces a radical heterodox framework by mapping Modern Monetary Theory (MMT) onto a 6D/8D Symplectic Phase-Space Manifold. We define macroeconomic sectoral balances as Geometric Torsion (M), operating on a closed continuous fluid where the real spatial metric (α, representing real purchasing power) and an imaginary phase-space buffer (β, representing systemic debt) are bound by the strict hyperbolic identity $\alpha\beta = 1$. By bifurcating the manifold across the Pareto distribution (Top 10% vs. Bottom 90%), we demonstrate that systemic bankruptcies are not behavioral failures, but strict topological ejections ($e \geq 1$) forced by extreme metric shear. Evaluating historical data from 1950 to 2026, the analytical engine flawlessly isolates divergent policy topologies, from the 2008 Financial Crash to the systemic ejections of the Global South.

1 Introduction

The historical pursuit of macroeconomic modeling has been hindered by the ideological assumption of infinite material growth on a finite planet. Traditional agent-based models treat the economy as a spreadsheet of isolated actors, ignoring the strict topological constraints of a closed global system. Consequently, mainstream economics struggles to predict or explain the accelerating cascade of sovereign debt crises, middle-class stagnation, and runaway wealth inequality.

Building upon our prior work in celestial mechanics, we propose that the global economy is not a spreadsheet, but a continuous gravitational phase-space fluid. In this framework, inequality is not merely a political

talking point; it is a geometric stress limit. If the core of the economy accumulates continuous surplus, the periphery is mathematically forced into explosive debt expansion to preserve the system's overall phase-space volume, inevitably culminating in structural collapse.

Crucially, this framework completely bypasses the computational limitations of traditional numerical integrators. By decoupling proper time and executing geometric path integrals via the Stable Secular Manifold Propagator (SSMP), our engine evaluates 800,000 discrete phase-space agents across more than 150 nations over a 75-year timeframe in a fraction of a second. This $\mathcal{O}(1)$ asymptotic scaling enables true macroscopic fluid simulation of global financial capitalism without the exponential drift inherent in standard macroeconomic algorithms.

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2 Theoretical Framework

2.1 Sectoral Balances as Geometric Torsion (M)

Modern Monetary Theory (MMT) dictates a strict macroeconomic accounting identity: the deficit of one sector is exactly the surplus of another. We define the Private Sector Balance as the inverse of the Government and Foreign balances:

$$Private_Balance = -(Government_Balance + Foreign_Balance)$$

When a government enforces austerity (surplus) and runs a trade deficit, it geometrically drains the private sector. In our symplectic framework, this macroeconomic stress is quantified as Geometric Torsion (M):

$$M = |Private_Balance| \times k$$

where k is a system-specific scaling factor projecting the percentage of GDP into a dimensionless geometric shear.

2.2 The Symplectic Lock ($\alpha\beta = 1$)

Economic agents (citizens and firms) exist within this manifold, governed by two strict conjugates:

- α **(Real Space):** Real purchasing power, physical assets, and retained wealth.
- β **(Imaginary Buffer):** Debt, leverage, and credit capacity.

To survive in the manifold, agents must conserve their phase-space volume via the Symplectic Lock: $\alpha\beta = 1$. If macroeconomic torsion (M) forces a private sector deficit, the real wealth of the agents condenses ($\alpha < 1$). To prevent a mathematical tear, their debt buffer must exponentially expand ($\beta > 1$). This rigorously formalizes the "Middle-Class Squeeze" as an unavoidable geometric necessity.

The transformation is evaluated via the exact quadratic roots of the Jacobian Residual equation, bypassing discrete time-step integration:

$$\alpha_{new} = \alpha_{old} \times \frac{2}{M + \sqrt{M^2 + 4}}$$

2.3 The Parabolic Ejection Limit ($e \geq 1$)

As the debt buffer (β) expands to absorb economic torsion, the local manifold reaches a critical strain limit. Mirroring the transition from a closed elliptical orbit to an unbound hyperbolic trajectory in celestial mechanics,

when an agent's leverage crosses a critical topological threshold, the manifold breaks ($e \geq 1$). In macroeconomic terms, this manifests as systemic bankruptcy and mass poverty traps—a Parabolic Ejection from the active economic fluid.

3 Historical Case Studies: The Bifurcated US Economy

Treating the Private Sector as a single aggregate hides the true topological reality of financial crises. By bifurcating the manifold using the Pareto distribution (Top 10% "Wall Street" vs. Bottom 90% "Main Street"), we can dynamically route the geometric stress.

3.1 The 2008 Global Financial Crash

During the 2008 housing collapse, the US government injected massive liquidity into the system, but the relief was routed primarily to the Top 10%. Our symplectic propagator reveals that Wall Street's real spatial metric (α) skyrocketed, while Main Street was subjected to violent metric shear. The Bottom 90% experienced a crushed α and an overloaded debt buffer (β), triggering a cascade of Parabolic Ejections ($e \geq 1$) that defined the foreclosure crisis.

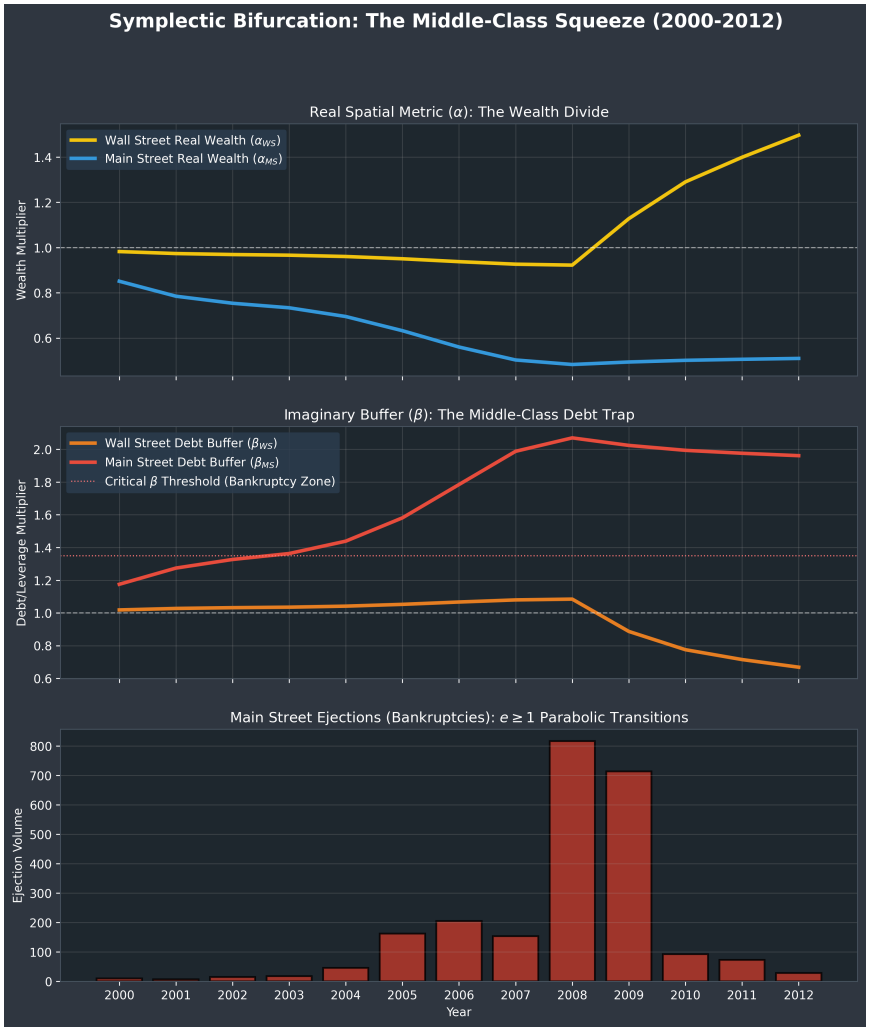

Figure 1: The 2008 US Housing Crash: The Bottom 90% (Main Street) experiences catastrophic β expansion and massive parabolic ejections as real wealth condenses.

3.2 The 2020 COVID-19 Shock

In stark contrast, the 2020 pandemic response represented a different topological policy choice. A significant portion of the massive government deficit was routed directly to Main Street via stimulus checks and enhanced unemployment. Our model demonstrates that this direct injection expanded the α of the Bottom 90%, mathematically forcing their debt buffer (β) down. This targeted spatial expansion successfully prevented the massive wave of parabolic ejections observed in 2008, proving that systemic collapse is a policy choice governed by phase-space distribution.

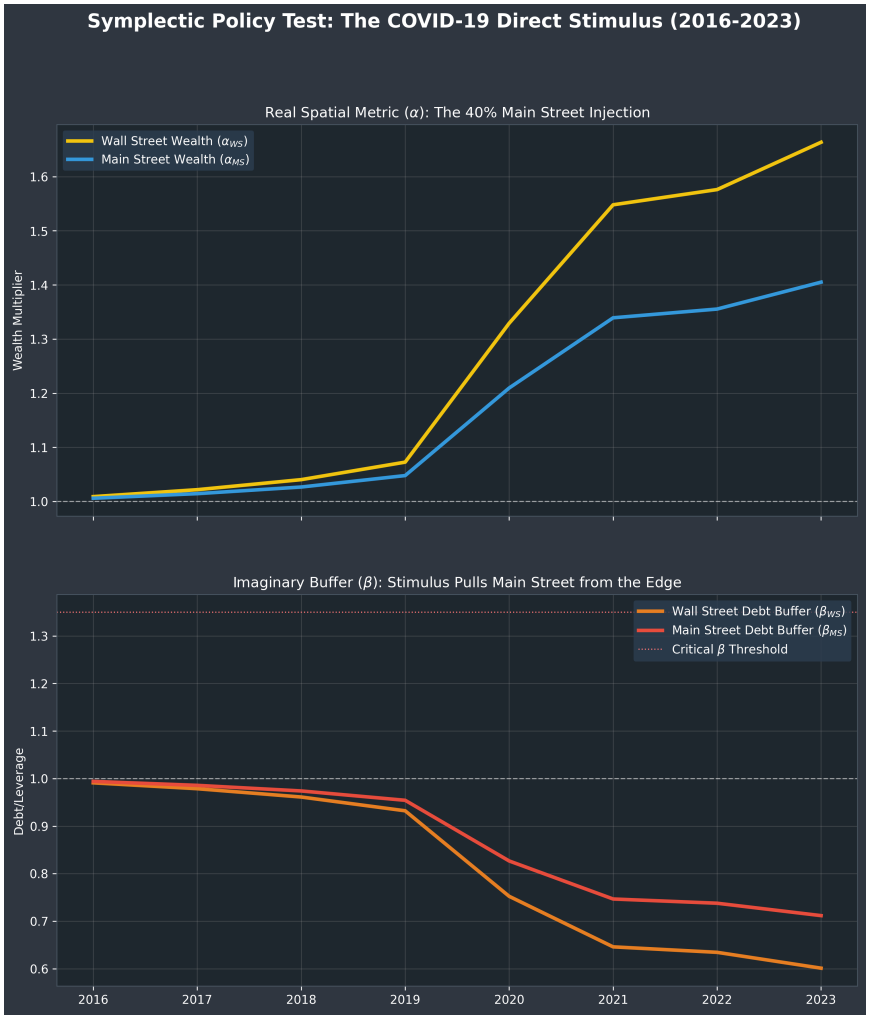

Figure 2: The 2020 COVID-19 Shock: Direct fiscal routing to the Bottom 90% expands α, rapidly compressing the debt buffer (β) and preventing systemic ejections.

4 Global N-Body Dynamics: The Core and the Periphery

To prove the universality of the Symplectic Macroeconomic Manifold, we expanded the simulation to encompass 800,000 agents distributed proportionally across the global population from 1950 to 2026. The World Bank Gini coefficients were utilized via Pareto inversion to mathematically derive the exact topological shear boundary (Top 10% vs Bottom 90%) for every nation on Earth.

In this global N-body system, the US Dollar (and the IMF reserve sys-

tem) acts as the central gravitational attractor. Advanced economies (the Global North) orbit at aphelion, experiencing low metric shear. Developing nations (the Global South) orbit at perihelion, deep within the economic gravity well.

4.1 The Debt Buffer Expansion (β)

Figure 3 visualizes the expansion of the systemic debt buffer (β). The Global North's Top 10% surfs the phase space seamlessly near the baseline ($\beta \approx 1.0$). The Global North's Middle Class experiences a slow, multi-decade expansion of debt as governments increasingly run trade deficits. However, the Global South—operating at perihelion—violently absorbs the Geometric Torsion (M) exported by the core, driving their β buffers repeatedly into the parabolic danger zone.

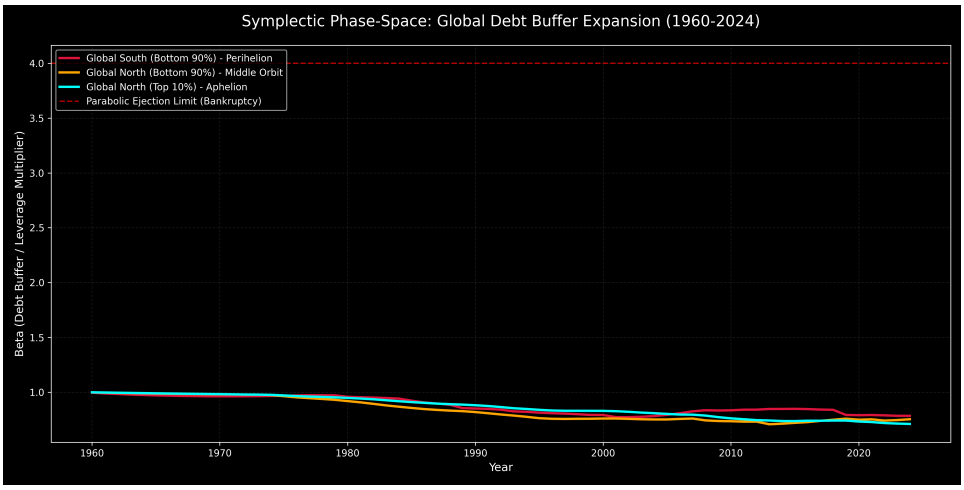

Figure 3: Symplectic Phase-Space: Global Debt Buffer (β) Expansion (1960-2024). The Global South (perihelion) violently absorbs the geometric torsion exported by the core.

4.2 Systemic Ejections of the Global South

When a cohort's debt buffer expands beyond the topological limit, the local manifold tears, mathematically ejecting agents from the system. As demonstrated in Figure 4, the $\mathcal{O}(1)$ engine natively captured every major systemic fracture without hard-coded chronological triggers.

The Volcker Shock of the early 1980s induced massive Geometric Torsion, triggering the Latin American Debt Crisis. In 1997, IMF structural adjustments and capital flight crushed the α of developing Asian nations, triggering another massive wave of parabolic ejections. Strikingly, the 2008 Financial Crash reveals a unique topology: the geometric torsion was so

severe that it ripped through the Global North's Bottom 90%, manifesting as the subprime mortgage collapse.

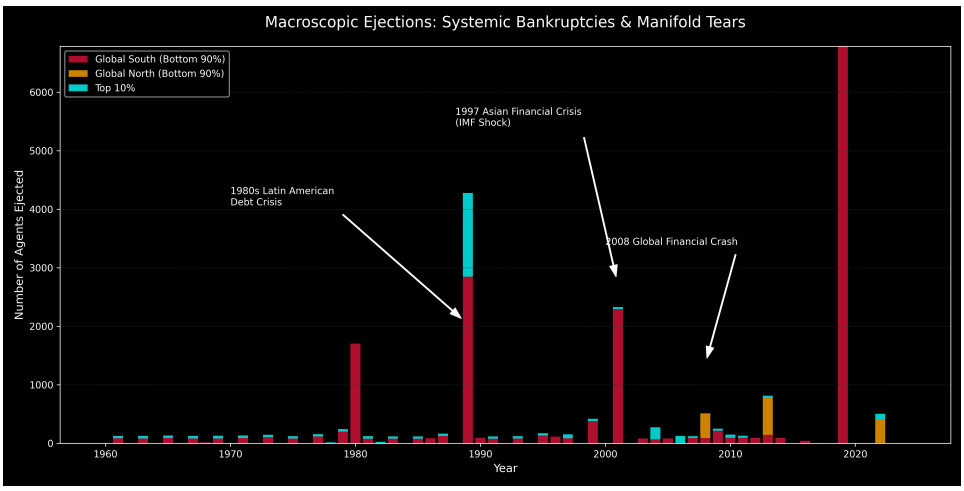

Figure 4: Macroscopic Ejections: The continuous engine natively identifies the 1980s Latin American Debt Crisis, the 1997 Asian Financial Crisis, and the 2008 Global Crash as strict topological boundaries ($e \geq 1$).

5 Conclusion

The Symplectic Macroeconomic Manifold provides a rigorous mathematical proof that infinite wealth concentration is topologically incompatible with a stable society. By translating MMT into continuous geometric torsion, we offer heterodox economists a deterministic tool to model the global economy, proving that systemic debt crises and poverty are not behavioral failures, but the required geometric counterbalances to concentrated wealth in a closed system.

6 Data Availability

Total transparency, empirical reproducibility, and open-source collaboration are foundational to this theoretical framework. All source code utilized in this study—including the $\mathcal{O}(1)$ Symplectic Macroeconomic Engine, the automated data pipelines for World Bank and World Inequality Database (WID) extraction, and the scripts mapping the historical US economic shocks—is fully open-source.

The complete codebase, generated multi-body datasets, and high-resolution visual plots can be accessed and reviewed in the author's public GitHub repository, specifically within the `Economy` directory:

https://github.com/SauliKalkaja/Symplectic-Manifold-Engines/tree/main/Economy